

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 1 – Pseudocode Design

Course Code:	DSA0405	Course Title:	FUNDAMENTALS OF DATA SCIENCE
CO Assessed:	CO5 –Pseudocode Design	Assessment:	Assessment Tool 1– Pseudocode Design
Weightage:	40% of CIA	Total Marks:	100
CO4 Statement:	Analyze and extract image features using detection and matching techniques for effective image representation and comparison. (BL4)		
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Section / Batch:	CSE – AI	Date:	25/08/26

Code of Conduct

I Mohamed Bazith M (192472052) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: Bazith

Criteria	Weight age	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning	Marks
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding; problem not identified correctly	
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts	
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning	

Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided	
Clarity	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand	

1. K-Means Clustering Algorithm

Problem Statement

K-Means clustering is an unsupervised machine learning algorithm used to divide a collection of data points into a specified number of groups called clusters. The main objective of the algorithm is to place similar observations in the same cluster and different observations in different clusters. Initially, a set of centroids is selected, where each centroid represents the center of a cluster. Each observation is then assigned to the cluster with the nearest centroid based on a distance measure such as Euclidean distance. After assigning all observations, the centroid of each cluster is recalculated using the mean of the observations belonging to that cluster. This process of assigning observations and updating centroids is repeated until the centroids no longer change significantly. The final result is a set of well-defined clusters with their corresponding centroid values.

Algorithm

1. Start the program and read the input dataset.
2. Select the required number of clusters, K.
3. Initialize K centroids randomly from the dataset.
4. Calculate the distance between every observation and each centroid.
5. Assign each observation to the cluster with the nearest centroid.
6. Calculate the new centroid of each cluster by finding the mean of its observations.
7. Compare the new centroids with the previous centroids.
8. Repeat the assignment and centroid update steps until convergence is reached.
9. Display the final clusters and their centroid values.
10. Stop the program.

Pseudocode

BEGIN

```
INPUT dataset D
INPUT number of clusters K
INITIALIZE K centroids randomly
REPEAT
  FOR each observation x in dataset D DO
    CALCULATE distance between x and each centroid
    ASSIGN x to the cluster with minimum distance
  END FOR
  FOR each cluster j from 1 to K DO
    CALCULATE new centroid as the mean of all
      observations assigned to cluster j
  END FOR
UNTIL centroids do not change
DISPLAY final clusters
DISPLAY final centroids
END
```

Output

K-Means Clustering Result

Cluster 1: Data points assigned to Cluster 1

Cluster 2: Data points assigned to Cluster 2

Cluster K: Data points assigned to Cluster K

Final Centroids:

Centroid 1 = (value1, value2, ...)

Centroid 2 = (value1, value2, ...)

Centroid K = (value1, value2, ...)

2. K-Means Clustering to Group Customers into Three Clusters

Problem Statement

Customer segmentation is an important application of K-Means clustering in data science and business analytics. In this problem, customers are grouped into three different clusters based on their annual income and spending score. Annual income represents the financial earning capacity of a customer, while the spending score represents the customer's purchasing behavior. By applying K-Means clustering with K equal to three, customers having similar income and spending patterns can be grouped together automatically. The algorithm begins by selecting three initial centroids and calculating the distance of every customer from these centroids. Each customer is assigned to the nearest cluster, and the centroids are recalculated based on the mean values of the customers in each cluster. The process continues until the cluster assignments become stable. The final three clusters can help identify different customer groups for better marketing and business decisions.

Algorithm

1. Start the program and read the customer dataset.
2. Extract Annual Income and Spending Score as input features.
3. Set the number of clusters K to 3.
4. Initialize three centroids for the three clusters.
5. Calculate the distance of each customer from all three centroids.
6. Assign each customer to the nearest centroid.
7. Calculate the new centroid for each cluster using mean values.
8. Repeat the process until the centroids do not change.
9. Label the customers according to Cluster 1, Cluster 2, and Cluster 3.
10. Display the customers and final cluster centroids.

Pseudocode

BEGIN

INPUT customer dataset

SELECT features:

 Annual Income

 Spending Score

SET $K = 3$

INITIALIZE 3 centroids

REPEAT

 FOR each customer c DO

 CALCULATE distance from c to Centroid 1

 CALCULATE distance from c to Centroid 2

 CALCULATE distance from c to Centroid 3

 ASSIGN customer c to nearest centroid

 END FOR

 CALCULATE new centroid of Cluster 1

 CALCULATE new centroid of Cluster 2

 CALCULATE new centroid of Cluster 3

UNTIL centroids remain unchanged

DISPLAY Cluster 1 customers

DISPLAY Cluster 2 customers

DISPLAY Cluster 3 customers

DISPLAY final centroids

END

Output

Customer Clustering Result

Cluster 1:

Customers with similar annual income and spending score.

Cluster 2:

Customers with similar annual income and spending score.

Cluster 3:

Customers with similar annual income and spending score.

3. K-Means Clustering Using the Elbow Method

Problem Statement

Selecting the correct number of clusters is an important step before applying the K-Means clustering algorithm. If the number of clusters is too small, different groups of data may be incorrectly combined together. If the number of clusters is too large, the data may be divided into unnecessary groups. The Elbow Method is commonly used to determine a suitable number of clusters. In this method, K-Means clustering is performed for different values of K, and the Within-Cluster Sum of Squares (WCSS) is calculated for each value. A graph is then plotted with the number of clusters on the X-axis and WCSS on the Y-axis. The point where the reduction in WCSS starts becoming slower is called the elbow point. This elbow point indicates a suitable value of K, which is then used to perform the final K-Means clustering and obtain meaningful groups from the dataset.

Algorithm

1. Start the program and read the input dataset.
2. Select a range of possible cluster values, such as $K = 1$ to 10.
3. Apply K-Means clustering for each value of K.

4. Calculate the WCSS value for every K.
5. Store the WCSS values for all cluster numbers.
6. Plot the Elbow graph using K and WCSS values.
7. Identify the elbow point in the graph.
8. Select the corresponding number of clusters as the optimal K.
9. Apply K-Means again using the selected K value.
10. Display the final clusters and centroid values.

Pseudocode

BEGIN

INPUT dataset D

FOR K = 1 to 10 DO

 APPLY K-Means clustering with K clusters

 CALCULATE WCSS

 STORE K and WCSS value

END FOR

PLOT graph:

 X-axis = Number of Clusters

 Y-axis = WCSS

IDENTIFY elbow point

SET optimal_K = elbow point value

APPLY K-Means clustering using optimal_K

DISPLAY final clusters

DISPLAY final centroids

END

Output

Elbow Method Result

Number of Clusters WCSS

1	High value
2	Lower value
3	Lower value
4	Lower value

Elbow Point Identified: K = 3

Optimal Number of Clusters = 3

Final K-Means clustering is performed successfully
using 3 clusters.

4. Evaluation Metrics: TP, TN, FP, FN, Accuracy, Precision, Recall and F1-Score

Problem Statement

Evaluation metrics are used to measure the performance of a machine learning classification model. For a binary classification problem, the actual class labels are compared with the predicted class labels generated by the model. Based on this comparison, four important values are calculated: True Positive (TP), True Negative (TN), False Positive (FP), and False Negative (FN). A True Positive occurs when the model correctly predicts the positive class, while a True Negative occurs when it correctly predicts the negative class. False Positive and False Negative represent incorrect predictions made by the model. These four values form the basis for calculating performance measures such as accuracy, precision, recall, and F1-score. Accuracy measures the overall correctness of the model, precision measures the correctness of positive predictions, recall measures the ability to identify actual positive cases, and F1-score provides a balanced measure of precision and recall.

Algorithm

1. Start the program and read actual and predicted class labels.
2. Initialize TP, TN, FP, and FN values to zero.
3. Compare each actual label with its corresponding predicted label.
4. Increase TP when both actual and predicted values are positive.
5. Increase TN when both actual and predicted values are negative.
6. Increase FP when the prediction is positive but the actual value is negative.
7. Increase FN when the prediction is negative but the actual value is positive.
8. Calculate accuracy using TP, TN, FP, and FN.
9. Calculate precision, recall, and F1-score.
10. Display all the evaluation metric values.

Pseudocode

BEGIN

INPUT actual labels

INPUT predicted labels

SET TP = 0

SET TN = 0

SET FP = 0

SET FN = 0

FOR each label i DO

 IF actual[i] = 1 AND predicted[i] = 1 THEN

 TP = TP + 1

 ELSE IF actual[i] = 0 AND predicted[i] = 0 THEN

 TN = TN + 1

 ELSE IF actual[i] = 0 AND predicted[i] = 1 THEN

 FP = FP + 1

 ELSE IF actual[i] = 1 AND predicted[i] = 0 THEN

 FN = FN + 1

```

    END IF
END FOR
CALCULATE Accuracy = (TP + TN) / (TP + TN + FP + FN)
CALCULATE Precision = TP / (TP + FP)
CALCULATE Recall = TP / (TP + FN)
CALCULATE F1-Score =
    2 * Precision * Recall / (Precision + Recall)
DISPLAY TP, TN, FP, FN
DISPLAY Accuracy
DISPLAY Precision
DISPLAY Recall
DISPLAY F1-Score
END

```

Output

Confusion Matrix Values

True Positive (TP) = 40

True Negative (TN) = 45

False Positive (FP) = 5

False Negative (FN) = 10

Accuracy = 85%

Precision = 88.89%

Recall = 80%

F1-Score = 84.21%

5. Model Evaluation and Selection

Problem Statement

In a data science problem, different machine learning algorithms may produce different results when applied to the same dataset. Therefore, it is necessary to evaluate and compare multiple models to select the most suitable one. In this problem, k-Nearest Neighbors (kNN), Decision Tree, Logistic Regression, and K-Means models are considered for evaluation. The dataset is first prepared and divided into suitable training and testing data where applicable. Supervised learning models such as kNN, Decision Tree, and Logistic Regression are evaluated using classification metrics such as accuracy, precision, recall, and F1-score. K-Means is an unsupervised clustering algorithm and can be evaluated using suitable clustering measures such as the Silhouette Score or WCSS. After calculating the evaluation results, the performance of all models is compared. The model with the most suitable evaluation results for the given data science problem is selected as the final model.

Algorithm

1. Start the program and read the input dataset.
2. Preprocess the data by handling missing values and selecting features.
3. Split the data into training and testing sets for classification models.
4. Train the kNN model and calculate its evaluation metrics.
5. Train the Decision Tree model and calculate its evaluation metrics.
6. Train the Logistic Regression model and calculate its evaluation metrics.
7. Apply K-Means clustering and calculate a suitable clustering score.
8. Compare the performance results of all the models.
9. Select the model that provides the most suitable result for the problem.
10. Display the evaluation results and the selected model.

Pseudocode

```
BEGIN
INPUT dataset D
PREPROCESS dataset D
SPLIT dataset into training data and testing data
TRAIN kNN model
PREDICT test data using kNN
CALCULATE Accuracy, Precision, Recall and F1-Score
TRAIN Decision Tree model
PREDICT test data using Decision Tree
CALCULATE Accuracy, Precision, Recall and F1-Score
TRAIN Logistic Regression model
PREDICT test data using Logistic Regression
CALCULATE Accuracy, Precision, Recall and F1-Score
APPLY K-Means clustering
CALCULATE clustering evaluation score
COMPARE all model evaluation results
SELECT the most suitable model
DISPLAY evaluation results
DISPLAY selected best model
END
```

Output

Model Evaluation Result

Model	Evaluation Result
kNN	Accuracy = 88%
Decision Tree	Accuracy = 91%
Logistic Regression	Accuracy = 89%

K-Means Silhouette Score = 0.72

Best Suitable Model:

Decision Tree

The model is selected based on the highest and most appropriate evaluation performance for the given problem.